Automotive Design 🚙

Brownie Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

Every car starts with a QUESTION

“Who is this car for?”

Today you design a car for a real driver.

Cars have a job and a driver 👩‍🚒

Look at these vehicles. Who drives it? What must it do?

- 🚒 Fire truck → needs a big ladder
- 🍦 Ice-cream truck → needs a freezer
- Race car → needs to go FAST
- 🚐 Minivan → needs lots of seats

Cars are designed around their driver's needs.

Meet your customer 🎴

Draw a **customer card**. This is who your car is for!

- A farmer 🌾
- A dog-walker 🐕
- A pizza deliverer 🍕
- A family with three kids 👨‍👩‍👧‍👧

Fill in your design sheet ✏

1. **My driver** is…
2. **What they need** is…
3. **My car's special feature** is…

Designing for a **real customer** — not just "a cool car" — is the whole badge.

Sketch FIRST, then build ✍

Designers always draw before they build.

1. **Sketch** your car on paper
2. Then build a cardboard **concept model**
3. Make the wheels **turn**!

It doesn't have to drive. It has to **show your idea**.

The Design Show 🎤

Show us your car in ONE sentence:

“This is for a , *so it has* !”

Example: “This is for a dog-walker, so it has a comfy pet seat!”

Gallery walk 🖼️

Walk around. Look at everyone’s cars.

Give **one kind comment** to a designer.

Robot moment 

Our team designs our robot around the **year's challenge**.

We decide what it needs to **do** before we decide what it **looks like** — just like designing a car for a driver.

You did it! 🎉

Today you: - ✅ Learned that vehicles are designed around a driver's needs - ✅ Described your customer and what their car must do - ✅ Sketched your design before building - ✅ Built a concept model with a special feature

You are a car designer now!

Keep going at home! 

- Look at cars on your street: **who is each one for?**
- Sketch a car for someone in your family
- Ask your troop leader about the next Automotive badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.